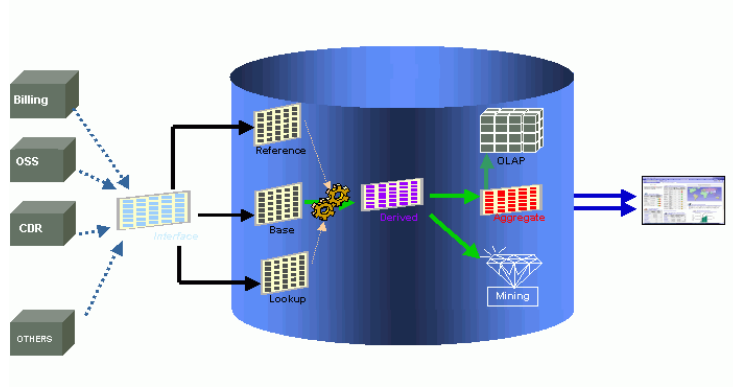




*from Oracle

CSCE 411/811

Introduction to Data Modeling

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College of Engineering
SCHOOL OF COMPUTING

Example

- Bad design increases your business cost!

Table 1.2 A badly-designed relational database table.

Movie:

<i>movieNr</i>	<i>movieTitle</i>	<i>releaseYr</i>	<i>director</i>	<i>star</i>
1	Cosmology	2006	Lee Lafferty	?
2	Kung Fu Hustle	2004	Stephen Chow	Stephen Chow
3	The Secret Garden	1987	Alan Grint	Gennie James
3	The Secret Garden	1987	Alan Grint	Barret Oliver
4	The Secret Garden	1993	Agnieszka Holland	Kate Maberly
4	The Secret Garden	1993	Agnieszka Holland	Heydon Prowse
5	The DaVinci Code	2006	Ron Howard	Tom Hanks
5	The DaVinci Code	2006	Ron Howard	Ian McKellen
5	The DaVinci Code	2006	Ron Howard	Audrey Tautou

Re-modeling (normalizing)

- Remove redundancy, add dependency

Movie (movieNr, movieTitle, releaseYear, director)

Starred (movieNr, star)



Movie:

<i>movieNr</i>	<i>movieTitle</i>	<i>releaseYear</i>	<i>director</i>
1	Cosmology	2006	Lee Lafferty
2	Kung Fu Hustle	2004	Stephen Chow
3	The Secret Garden	1987	Alan Grint
4	The Secret Garden	1993	Agnieszka Holland
5	The DaVinci Code	2006	Ron Howard

Starred:

<i>movieNr</i>	<i>star</i>
2	Stephen Chow
3	Gennie James
3	Barret Oliver
4	Kate Maberly
4	Heydon Prowse
5	Tom Hanks
5	Ian McKellen
5	Audrey Tautou

Data Modeling



Hierarchical
Data modeling technique

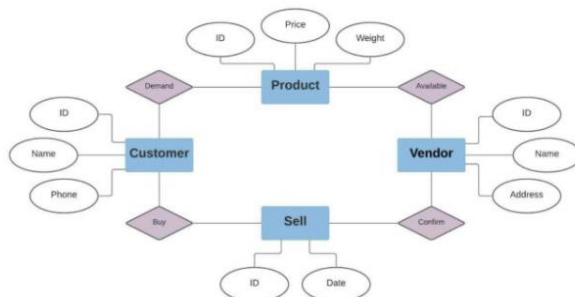


Relational
Data modeling technique

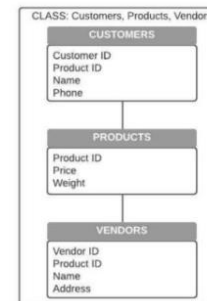


Network
Data modeling technique

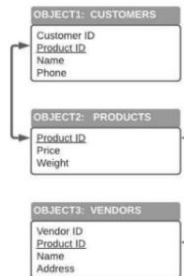
Data Modeling Techniques



Entity-relationship
Data modeling technique



Object-oriented
Data modeling technique



Example

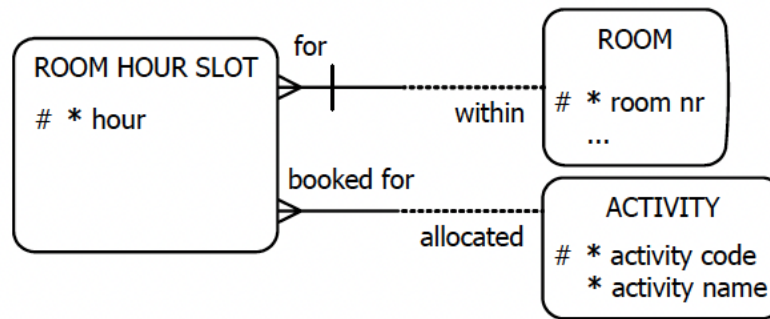
- Student class enrollment system
- What are the users?
- What are typical user cases?
- What are the essential data?
- How to organize the data to support user cases?
- What shall the data representation and relationships?
- What is the DBMS needed?

Role Communication

- Domain experts
- Modeler
- Users

Entity Relationship Modeling

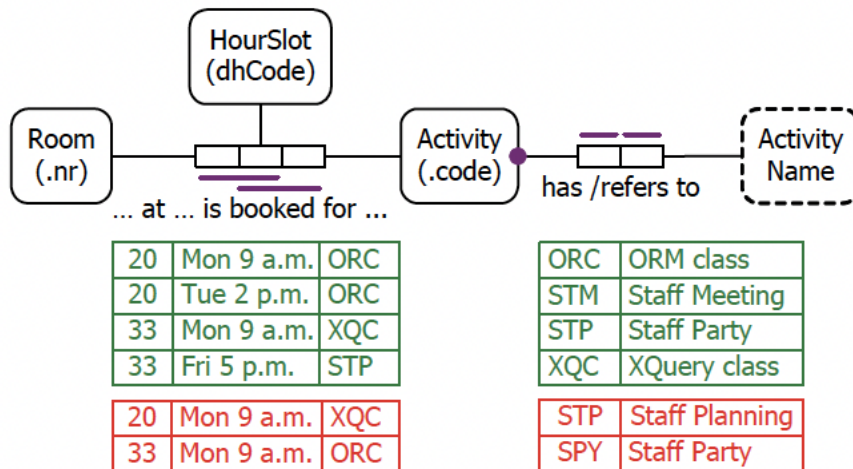
- # is the key attribute
- * cannot be empty



Chen model 1969

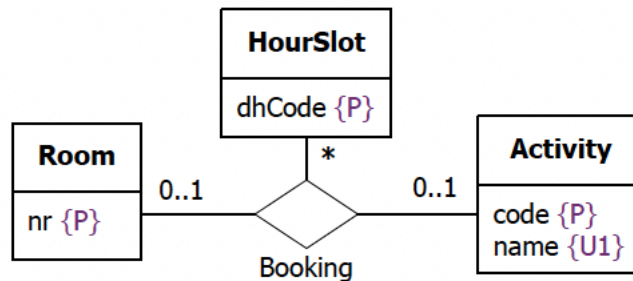
Object Role Modeling

- Red means conflicts
- Room&hourslot with at most one activity
- Hourslot&activity with at most one room
- Activity is an entity; ActivityName is a value.



Object Oriented Modeling

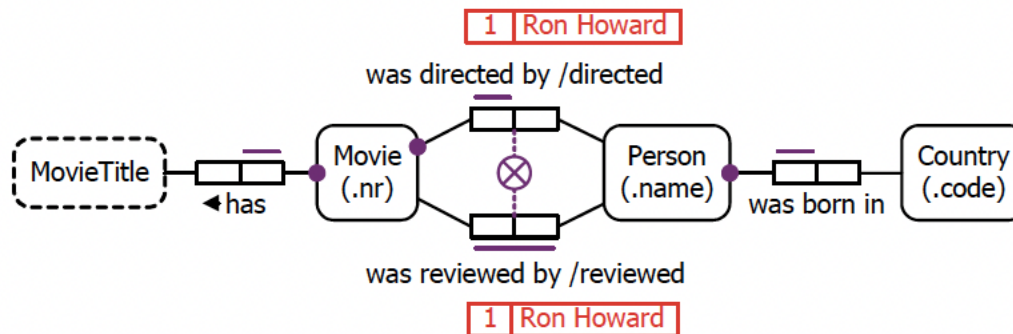
- UML class diagram



U1: additional key

ORM and UML

- UML cannot represent exclusive situations for two relationships



Movie	movieDirected	director	Person
nr {P}	*	1	name {P}
title	*	*	birthCountry
	movieReviewed	reviewer	

ORM and UML

- ORM is closer to natural language representation.
- There are many data modeling strategies.
- UML can be used on both programming and database design.
- ER is good for structured data.
- Constraints are hard to present in the diagram.